Blending Immersive Gameplay with Intense Exercise using Asynchronous Exergaming

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ABSTRACT

Exergames are video games that encourage players to do physical exercises (e.g. yoga, aerobics). While exergames provide some motivational and fitness benefits, they are limited by shallow gameplay and low-intensity exercise. This occurs because, in most existing exergames (e.g. Wii Fit), the user plays the game and exercises simultaneously. By developing exergames where gameplay and exercise occur at separate times, we can motivate high-intensity exercise with high-immersion gameplay. We therefore developed such an *asynchronous* exergaming by modifying the popular game *Skyrim* to use exercise data from the fitness tracking website *Exercise.com*. This paper describes the design and implementation of the Skyrim exergaming mod.

CCS Concepts

• **Applied computing**→Computers in other domains→Personal computers and PC applications→Computer games

Keywords

Exergame; Game Design; Role-Playing Game; Prototype; Proof-of-Concept

1. INTRODUCTION AND RELATED WORK

Obesity is a serious problem—arguably an epidemic. Here in New Zealand, two thirds of adults and an increasing proportion of children are either obese or overweight [14]. Two thirds of New Zealanders also play video games [3].

Video games are sometimes implicated in the obesity epidemic because they may encourage a sedentary lifestyle. However, motivation is crucial for successful exercise programs [18] and video games are associated with heightened motivation [15]—not to mention heightened engagement [6], immersion [21] mindfulness [9] and flow [4, 5, 22] – a mental state characterized by focus, enjoyment and strong performance [7].

It seems plausible, then, that video games could be designed to encourage an active lifestyle. Since physical exercise helps obese and overweight people lose weight and is associated with diverse health improvements [20], such games could have many benefits. Several such exercise games, or *exergames*, have consequently been developed, beginning with Foot Craz for the Atari and Family Trainer for Nintendo in the late 1980's.

Today, exergames can be divided into three broad categories:

1. *Living-room exergames* such as Dance Dance Revolution and Wii Fit involve performing physical exercise, including dancing, aerobics and yoga, typically while at home in front of one's television.
2. *Cardio machine exergames* such as the Fish Game on the Concept2 rowing machine [11] are played while using specific training apparatus. Some use virtual reality.
3. *Mobile exergames* use accelerometers, global positioning systems or similar technologies to track player movement and affect gameplay. In *Zombies, Run!*, for example, the runner/player must speed up when chased by zombies.

Each category faces serious challenges. Even if living-room exergames delivered high-immersion gameplay (most do not), achieving high-intensity exercise in one's living room is extremely difficult. Mobile exergames, meanwhile, have the opposite problem—it is extremely difficult to make a high-immersion game that can be played (safely) while running, swimming, cycling or otherwise moving rapidly outdoors. Activity tracking applications including Fit Bit, Fitocracy and Nike+, which attempt to gamify (cf. [13]) the fitness logging process, similarly lack high-immersion gameplay. Unsurprisingly then, while current exergames stimulate some energy expenditure [2], "most commercial systems on the market today are insufficient in providing high levels of health benefits" [23]. Cardio machine exergames, in contrast, could potentially deliver both high-intensity exercise and high-immersion gameplay. However, these games require specialized hardware, which may be impractically expensive.

In summary, since motivation is positively related to not only immersion [15] but also exercise [17], and since health benefits are positively related to exercise intensity [20], high-immersion, high-intensity exergames should be more effective. However, combining immersion and intensity is difficult when players exercise *while* playing the game, as in most existing exergames. This suggests the following research question:

> ***Research Question:*** *Can an asynchronous exergame combine high-immersion gameplay and high-intensity exercise?*

Here, an *asynchronous exergame* is a video game that promotes exercise outside of play sessions. *Immersion* is the degree of a player's mental involvement in a game. *Intensity* is the degree of exertion (both muscular and cardiovascular) during exercise.

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GAS '16, May 16, 2016, Austin, TX, USA.
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ACM 978-1-4503-4160-8/16/05...$15.00
DOI: http://dx.doi.org/10.1145/2896958.2896959

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To investigate the above research question, this paper describes the design and implementation of a prototype asynchronous exergame (Section 2), and presents key lessons learned (Section 3). Section four concludes the paper with a broad agenda for future research.

2. DESIGN AND IMPLEMENTATION

2.1 Concept Inception and Prototype

The aim of this project is to create an asynchronous exergame—a virtual experience that encourages players to exercise in real life. The basic strategy is to combine a fitness tracking application with a video game such that fitness data has interesting in-game consequences. Our design was informed by contemporary game design theory, especially Schell's lenses [19] and the Unified Theory of Digital Games [16].

Since developing a good, immersive game from scratch is expensive and time consuming, we decided to modify an existing game. We chose the role-playing game *The Elder Scrolls V: Skyrim* for the following three reasons.

1. It is well-designed and very popular, having sold over 20 million copies [8].
2. With over 300 hours of content [10] and a strong progression aesthetic [12], it could support several months of physical training.
3. With a built in modification kit (called *Creation Kit*) and vibrant modding community, Skyrim is easy to modify. Furthermore, its three character attributes map well into fitness data (see below).

Skyrim is a single-player western role playing game developed by Bethesda for the PC, Playstation 3 and Xbox 360 (Figure 1). It has a medieval fantasy setting including dragons and magic. The player character in Skyrim has three main attributes: Health, Stamina, and Magicka, along with 18 different skills (e.g. Archery, Illusion, Pickpocketing). The players earn experience by completing quests and defeating enemies. When enough experience is gained, the player levels up and can choose attributes and skills to increase.

Figure 1 - Skyrim gameplay

The PC version of Skyrim ships with a modification development kit provided, called Creation Kit (Figure 2). Creation Kit is a platform that allows players to alter the game, for example, by creating new quests, non-player characters (NPCs) and items. It uses a proprietary scripting language called Papyrus as well as a graphical interface creation system. This has led to many publically available mods including improved texture packs, new quest chains, and new armour models. The modding community had created many helpful resources and documentation to assist modders. There are also forums and IRC channels available for Creation Kit modding support.

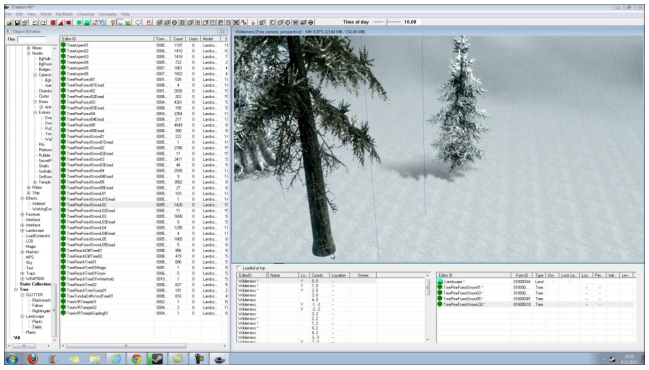

Figure 2. Skyrim Creation Kit

Similarly, after evaluating various fitness tracking applications, we chose Exercise.com (Figure 3) for the following three reasons.

1. It allows users to track a wide variety of exercises including strength, cardio, and sport training.
2. It has good cross-platform compatibility with free web, Android and iPhone applications.
3. It has a ready-made point system that may be used to convert exercise date into Skyrim experience points.

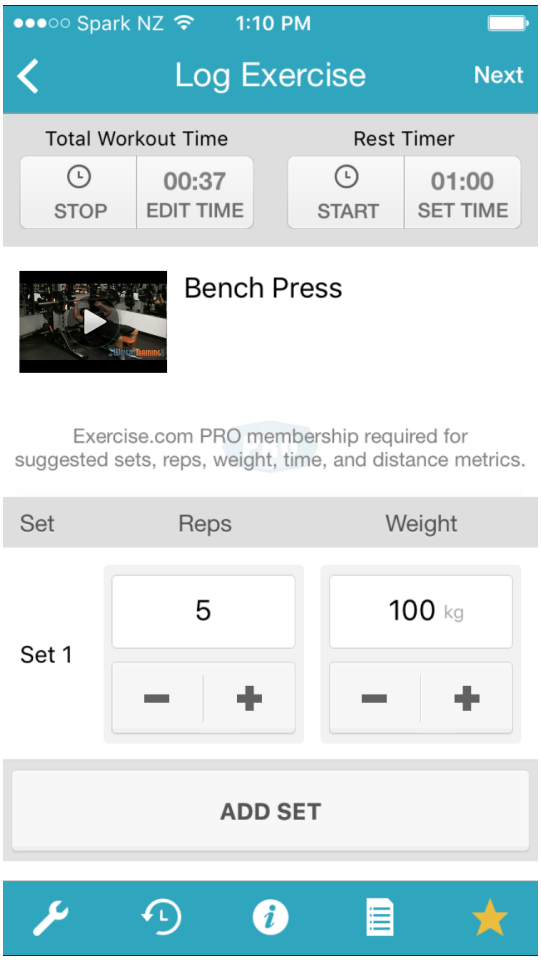

Figure 3 - Exercise.com logging interface (iPhone)

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While Exercise.com does not have a public API, it syncs with Runkeeper, which does. We therefore used the Runkeeper API to extract data from Exercise.com.

2.2 Proposed Exergame Model

There are four steps to our exergame model:

1. A user performs some physical activity. This could include any form of cardio, strength training or sports.
2. The user logs their activity using Exercise.com's logging platform.
3. When the game is opened, it fetches exercise data using the Runkeeper API.
4. The user is notified of his or her levelling status and resumes playing the game.

If the user has logged enough activity to level up they will be notified about how their attribute bonuses have been allocated. Otherwise, they will be notified about how far they are from levelling up (Figure 4).

Figure 4. Exergame model and exercise to attribute mappings

There are three character attributes in Skyrim: health, stamina and magicka. Health is how much damage a character can take before dying; stamina is the energy system used for running and performing special physical attacks; magicka is the energy system for performing magic attacks. We mapped types of physical activity into game attributes—cardio increased stamina, sports increase magicka and weight training increases health. The inherent progression of game difficulty creates an incentive for players to continue to participate in physical activity in order to progress with the game (Figure 5).

Figure 5. Exergame model and exercise to attribute mappings

2.3 Architecture

Implementation was split into two key components. The mod was developed using Creation Kit, and a Datasync client was developed using Python that would fetch data from Runkeeper.

The architecture of the mod comprises of two key components: a Datasync client and the Skyrim mod (Figure 6). The DataSync client first pulls user data that is logged through Exercise.com using the Runkeeper API. JSON objects are received from Runkeeper API request and the DataSync client sorts through these objects, extracts the relevant fields from the activities, translates these values into points and stores these points in a file. The Skyrim mod then reads this file when a player loads their saved game, applies the in-game experience and awards a level up if enough experience has been earned, and then updates the data file as necessary. The updates to the data file are done to prevent players from being able to reload the game over and over again and have the same experience being reapplied. The Datasync client and the Skyrim mod are encapsulated within a custom launcher, which simplifies the starting of Skyrim for end users. The user simply clicks on the custom launcher to log into Runkeeper, sync fitness data and start Skyrim.

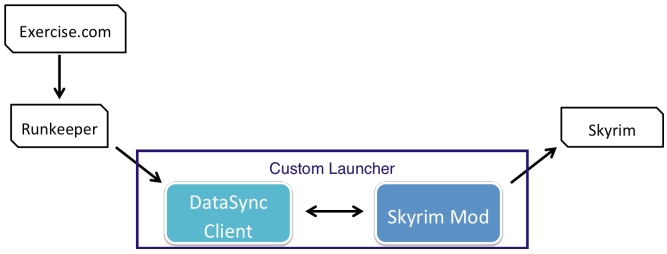

Figure 6. Mod architecture

2.4 Creation Kit Mod Implementation

Using Creation Kit, we created an NPC (Non-Player Character) named "Exer of Cise". Players can speak to this NPC in the game's first town to receive a quest. This quest is used to turn on exergame mode within Skyrim.

The exergame quest manages all exergame mode concerns such as leveling up the player, adjusting the experience requirements depending on the player and updating the exercise data file after experience points have been used.

2.4.1 Quest Stages

This quest is divided into six stages. Each stage is used to indicate a different state the player is currently in for the exergame mode.

1. Exergame mode is turned off and the player is essentially playing *Skyrim* unmodified.
2. The player first accepts the quest from Exer of Cise. A notification appears on the player's screen showing that they have accepted the exergame quest. Exer of Cise offers exergames-specific dialogue. The player exits the dialogue.
3. The player receives an immediate level-up for accepting the quest. This communicates that leveling up is not too difficult.
4. Data of the player's first workout is received and this triggers a level-up regardless of the content of the workout.
5. Stage five begins immediately after the player's first workout is processed and lasts for one week. Each time *Skyrim* starts, The DataSync client logs all exercise points earned and saves them into a field in the exercise data file. During this week, all

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players use receives level-ups based on a default experience curve. (This is necessary because insufficient data exists to support the tailored levelling used in stage six.)

6. Stage six comprises the rest of the game. Experience requirements are tailored to the player and increasing exercise magnitude, duration and intensity accelerates levelling (see below for details).

2.4.2 On load game level ups

Creation Kit uses event-driven programming and for the exergame quest, the OnPlayerLoadGame event is used as a trigger to check if a player has earned a level up. The exercise data file is read in and saves how much total strength exercise points, cardio exercise points and sport exercise points a player has. If a player has enough points in total to level up, the player will be leveled up.

During a level-up, the mod both increments the player's level and increases one or more attributes. This avoids Skyrim's default levelling system where players can delay levelling.

Each time the game loads, a message box is shown to the player that indicates whether they have leveled up or not. If the player has leveled up, they are congratulated and shown how much each attribute has increased by, otherwise the player is told how much more experience is required to level up.

The remaining experience points a player has will then be used to check whether the player has earned any more level ups to put onto the "Level Stack" (see below).

Lastly the OnPlayerLoadGame event method overwrites the data file so that players cannot load the same data file over and over to receive free level-ups.

2.4.3 Level Stack

The "Level Stack" prevents players from receiving multiple level-ups from a single data import. This stack works by awarding a player the first level they have enough experience for and then checking if they have enough experience for any subsequent levels. If the player does, they will be given an item called an "Orb of experience" for each level they have on the stack.

Every hour an update method performs a check to see see if the player has an "Orb of experience" in their inventory. If the player has one or more orbs, they will experience a level up similar to an OnPlayerLoadGame level up and one orb will be consumed. The orbs also allow the players to see pending level ups.

2.4.4 Level ups

All level-ups grant 10 attribute points distributed between the player's health, stamina, and magicka. The distribution of these points will depend on the exercises logged. For instance, if a player has only logged strength training since their last level-up, they will receive all 10 points to their health attribute. In contrast, if they have 70% of their points through cardio and 30% through sports, they will receive 7 points into stamina and 3 points into magicka.

2.5 Game Design

2.5.1 Points System

Exercise.com's point system returns the number of points earned in each logged training session. These number of points received will be in proportion to the quantity and quality of activity that is logged. The calculation of these points will depend on many factors including exercise intensity, exercise duration, exercise difficulty, type of equipment used and muscle groups involved. These calculations are very complex and have been developed by a large team of domain experts aided by community feedback. A point system is essential for our exergame model as it provides a metric that allows us to translate physical activity into in-game experience. Unfortunately, Exercise.com's point system is not accessible through the Runkeeper API or through any other means we could find. Since we could find no way to access Exercise.com data directly, we had to create our own rudimentary point system.

For the fitness activities (cardio and sports) each activity is assigned an intensity modifier between 1.5 and 2.5, depending on the intensity of the activity. This intensity modifier multiplied by the duration of the activity is how we determined the amount of points obtained from each fitness activity:

$$Points = IntensityModifier * Duration$$

$$where 1.5 \leq IntensityModifier \leq 2.5$$

For the purpose of our points and experience system, we required a baseline that all calculations are relative to. This is so that all points obtained across different types of activities and intensities can be scaled and fit together in the same experience tree. The baseline that was chosen is a fitness activity of intensity two and duration 90. An activity of moderate intensity done for 90 minutes is what we have interpreted as an "average" fitness activity session.

For the strength activities we used tonnage as the base metric for evaluating points obtained from each exercise. Tonnage is defined as (weight * sets * reps). Tonnage on its own cannot be directly translated into our point system as the numeric value would be too high. It was required that the tonnage values were scaled down in order to fit in with the fitness activities. This meant that an average strength training session should give a similar amount of points as an average fitness activity session. To do this we analysed the strength sessions of three trainees with average strength levels. The tonnage of each individual session was analysed and heavy sessions were compared with light sessions. Heuristic analysis was performed and it was found that a balancing constant of 29 would normalize the tonnage values to be able to fit alongside the fitness activities. The formula was then applied to an advanced trainee's sessions in order to see how it would scale appropriately. It was found that the increase in points was reasonable and was relatively proportional to an advanced fitness training session.

$$Points = \frac{Sets * Reps * Weight}{BalancingConstant}$$

$$\text{where BalancingConstant} = 29$$

2.5.2 Experience System

Many video games, like physical exercise [1], exhibit decreasing returns. Decreasing one's 10km run time from 60 minutes to 55 minutes typically takes less effort than decreasing it from 55 minutes to 50 minutes. Similarly, in most video games it takes less effort to get from level 5 to level 6 than to get from level 15 to level 16. This is part of the steadily increasing challenge curve used by games to keep players motivated. We therefore wanted to implement an experience curve with decreasing returns.

However, to reach a larger demographic, an exergame should scale across a wide range of fitness activity levels. The game should still be playable whether the player trains one per week and plays every day, or trains every day and plays once a week. Obviously this is

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extremely challenging. To proceed, we first work through an example for a typical player and then generalize the example to a broader range of players.

We divided the skeleton of the experience requirements for levelling into different brackets because linear progression in training eventually results in diminishing returns. Instead, piecewise functions were chosen such that the amount of time spent training would closely resemble the amount of time it took to level up in the original game. We used four brackets: (4-9) (10-19) (20-29) (30+). We made level 30 the lower bound for our last bracket because it was found that this is approximately the point where the player's character is potentially strong enough to complete any part of the game. Once the different experience brackets had been established we needed to figure out the experience requirements of each bracket. Note that levels 1-4 are free. The player begins on level 1, receives level 2 during the tutorial (before enabling exergaming mode), receives levels 3 upon accepting the exergaming quest and receives level 4 on first sync. The first bracket therefore comprises level 4-9.

To proceed, we make two assumptions:

1. An average training session lasts for 90 minutes. (Since the average intensity modifier for a fitness activity is 2, this gives 180 points per session).
2. An average player trains three times per week (giving 540 points per week).

While these numbers are much too high for sedentary or high-risk populations, initial testing is planned for low-risk, physically-active young adults.

To scale well with gameplay, we want players to reach approximately level 9 by the end of the first week; therefore, we make the total experience requirements for levels 4-9 approximately 540 points. Skyrim's levelling formula for the experience requirement at a fixed level is

$$XPRequiredForLevel_n = (Level_n - 1) * fXPLevelUpMult + fXPLevelUpBase$$

$$where \quad Level_n = desired \quad level \quad and \quad fXPLevelUpMult \quad and$$

$$fXPLevelUpBase \quad can \quad be \quad manipulated \quad programatically$$

Below, fXPLevelUpBase will be referred to as v1 and fXPLevelUpMult will be referred to as v2 for simplicity. V1 and v2 are the only variables that can be manipulated for adjusting the experience requirements, and as such these will be programmatically adjusted at each level bracket. We set v1=16 and v2=10 to get 546 points for week one.

For the second bracket (levels 10-19) we want the player to increase his or her activity level and complete the bracket in about two weeks. Assuming four, 90 minute sessions were week (or three, 120-minute sessions), we want approximately 1440 points in this bracket. Set v1=65 and v2=5 gives a close-enough 1425 points.

The same process is repeated for the third bracket. We want the third bracket to take 10 sessions, which takes 2.5 weeks at the rate of 4 sessions per week or remains at 2 weeks if the player increases activity to 5 sessions per week. Setting v1=105 and v2=3 which gives us 1815 points.

At level 30 and beyond we fix the experience requirements to increase linearly as this is the point where levelling up has lower importance in relation to the rest of the game. We have chosen 14 sessions per bracket of 10 levels. We chose v1=158 and v2=2 which results in 2520 points per 10 levels for levels above 30.

To generalize this to a range of players, we retain the first bracket to acquire initial training data in week one and then tailor the leveling curve based on previous data, according to the following formulas:

To reach a larger demographic, an exergame should scale across a wide range of fitness activity levels. The experience function described above cannot be fixed like the original game and instead we will need to tailor it to the user's fitness levels. The previously designed experience skeleton can be set up entirely with just the initial first week points i.e. 540 points. This means that if the same proportions of v1 and v2 are kept relative to the initial skeleton and only the points in the initial week are modified, the experience skeleton can be scaled based on the amount of activity done in week one. The following formulas are generated:

$$Bracket \quad 1 \quad variables : V1 = b1 * \frac{16}{540}, V2 = b1 * \frac{10}{540}$$

$$Bracket \quad 2 \quad variables : V1 = b2 * \frac{16}{540}, V2 = b2 * \frac{10}{540}$$

$$Bracket \quad 3 \quad variables : V1 = b3 * \frac{16}{540}, V2 = b3 * \frac{10}{540}$$

$$Bracket \quad 4 \quad variables : V1 = b4 * \frac{16}{540}, V2 = b4 * \frac{10}{540}$$

where:

$$BracketOnePoints \quad b1 = UserInputFromWeekOne$$

$$BracketTwoPoints \quad b2 = BracketOnePoints * 2.6667$$

$$BracketThreePoints \quad b3 = BracketTwoPoints * 1.25$$

$$BracketFourPoints \quad b4 = BracketThreePoints * 1.4$$

These formulas impose a soft cap of level 9 during the first week, that is, the curve scales such that no one could conceivably train hard enough to reach level 9. Capping progress in the first week is necessary to prevent players from levelling too quickly, which would make the game too easy and therefore boring. However, imposing a soft cap (where the player appears to continue advancing toward level 9) keeps the player motivated.

3. LESSONS LEARNED AND FUTURE WORK

We learned several important lessons by developing this prototype exergame:

1. Creation Kit does not fully support object-oriented practices or web services. Modders create their own object classes. In-game persistence and available data structures are also limited. We therefore had to save state using a custom inventory item and write a python client to retrieve the exercise data.
2. Early in the game, it is possible for the player to receive multiple level ups from one exercise session. Levelling too quickly can degrade game balance and feel overwhelming to a new player. To prevent this from happening, we implemented a level stack such that, if a player earns several level-ups at once, they receive the first immediately; however, subsequent level-ups are stored in inventory as "orbs of

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experience." Each hour, one Orb of Experience is automatically consumed, triggering a level-up.

3. Players may also decide to neglect one or two exercise categories; e.g., magicka is of little value to a non-magical character. This issue remains unresolved and we are waiting to see its effect in upcoming user studies.

4. The exercise point system is rudimentary and the intensity modifier only roughly captures trainee effort. While we hard-coded checks for low range-of-motion strength exercises (e.g. calf-raise, shrug), which would generate too many points otherwise, a more comprehensive approach is needed. Fortunately, Exercise.com seem willing to provide us with a way of accessing their point system in the future.

5. Only one save file is supported. This may frustrate some players and should be addressed in future work. is limited to working with only one save file. This is because all experience points a player has earned are applied to the character when the game is loaded. After points have been applied, they are removed from the data file to prevent players from re-importing old data to get free level ups.

3.1 The bigger picture

Different fitness tracking applications appeal to different kinds of trainees and different kinds of games appeal to different players. Players, moreover, tire of one game and move on to another. The broader aim of this project is therefore not simply to build a specific exergames but to create tools and technologies to facilitate exergaming modes in many games, which harvest data from many sources.

The current prototype suggests that such a system might have a three tier architecture (Figure 7):

1. Fitness data services including wearables (e.g. Fit Bit), manual tracking applications (e.g. Exercise.com) and automatic tracking applications (e.g. Runkeeper).

2. A data integration layer, which transforms raw fitness data into information that makes sense to and is more directly usable by game designers. This would include an exercise point system.

3. Video games with exergaming modes and stand-alone exergames.

4. CONCLUSION

In summary, this paper: proposes that asynchronous exergames (where gameplay and training occur separately) may be more effective and practical for motivating increased physical exercise; describes the implementation of and lessons learned from a prototype asynchronous exergame; and provides a simple framework for future development in this area.

Two avenues of future work are evident. While we are content to submit a simple description of the prototype to a workshop, we obviously need to empirically test the prototype's affect on actual trainees. This can be achieved through randomized controlled trials comparing a treatment involving the playing the prototype with a control such as a standard health and fitness intervention. Furthermore, a more comprehensive middleware component is needed to facilitate broader development of exergaming modes and integration of more data sources.

When building a commercial game, developers choose among a variety of modes, including online competitive multiplayer, offline co-operative multiplayer ("couch co-op") and variations enhanced difficulty ("hardcore mode"). The long-term objective of this research is to convince game publishers including Sony, Microsoft, EA, Nintendo and Valve that it will be easy, profitable and ethical to add exergame modes to many of their games, maximizing the positive effect on societal health and wellbeing. This will create a broad research program around how to implement exergaming modes in different kinds of games (e.g. puzzle, turn-based strategy, role-playing) and the relationship between numerous game features and effects on player motivation, health and fitness. Only when exergaming modes are as ubiquitous as online multiplayer will the full benefits of exergaming be realized.

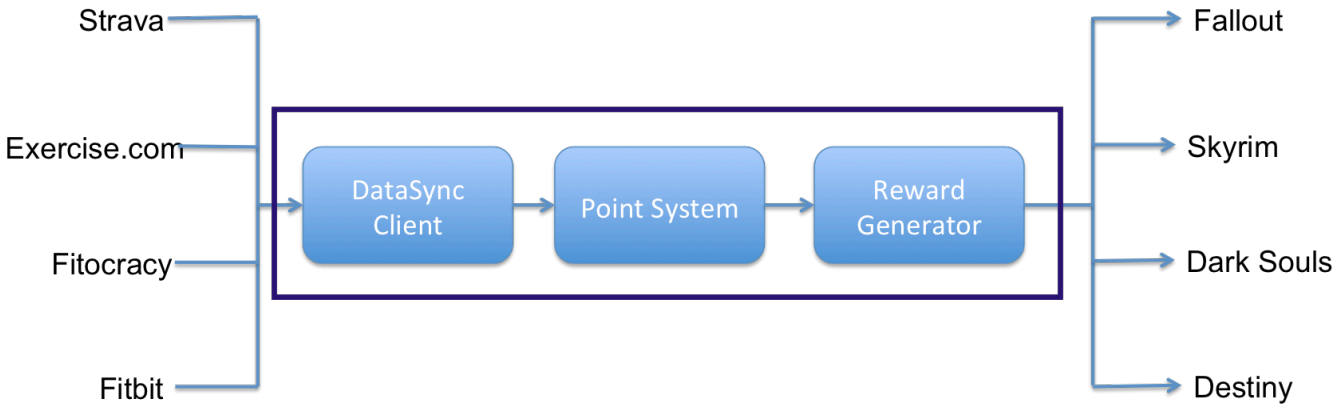

Figure 7. Future development

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